Rhapsody Protocol

Day 1: RBC Removal, PBMC Isolation, Rhapsody, cDNA/Template Switch

Stemcell EasySep RBC Depletion & PBMC CryoPres

1. **1-1.5mL for plasma, 1mL RBC depletion, rest for PBMC iso**
2. Plasma aliquot spin 10 mins 1,000 rcf, store in cryotube -80
3. RBC and PBMC aliquots add equal parts PBS+2%FBS
4. PBMC Isolation:
    a. 15mL Lymphopop to 50mL SepMate bottom
    b. Equal parts blood to PBS+2%FBS
    c. Gently add blood to SepMate, avoid mixing
    d. Spin down 10mins @ 1200g, move SN to new tube
    e. Save 1-2mL plasma for Ferhat Lab
    f. Spin down 5min 400 g, discard SN
    g. Wash x2: top off conical with PBS+FBS + spin
    h. Resus in FBS+10% DMSO
    i. Aliquot into 1.8mL cryovials → Mr. Frosty → LN2 tank → update spreadsheet
5. 2mL blood+PBS/DBS for RBC Depletion
    a. In round bottom 5mL flow tube add 48uL 0.5mM EDTA per 1mL of blood. Vortex 30 sec
    b. Add 50uL Depletion reagent and pipet mix
    c. Place on magnet without cap 5 mins
    d. Transfer SN from front of tube to new tube
    e. Repeat d. and e. for total of 2-3 washes
    f. Transfer to new tubes w/o Depletion reagent 5 mins
6. Spin down 5 mins 200g, carefully remove SN (platelets)
7. 10uL count, 1:1 dilution with EOSIN
8. Aliquot 1E6 cells into new 1.5mL tube each sample

AbSeq & Sample Tag Prep & Stain

1. Make Ab cocktail 1 (Abseq) and 2 (ST) for each sample
    a. Spin briefly and pipette mix, do not vortex!
    b. SKIP IF NOT MULTIPLEXING: Add 180uL of Ab cocktail-2 to 20uL of assigned Sample Tag
2. Spin cells down 400g 5mins 4C
3. Resuspend in 200uL Ab cocktail-1. **Incubate @RT 10mins**
4. Pipet mix to help visible pellet form. Spin 400g 5mins
5. Remove SN by inverting tube and brief block on paper towel
6. Resuspend in 200uL Ab cocktail-2. **Incubate on ice 30mins**

7. Add 1mL BD stain buffer and pipet mix
8. Spin 400g 5mins, remove SN. Do not vacuum, check pellet
9. Resuspend 500uL Staining Buffer
10. 10uL cell count, no dilution
11. Aliquot 150-200k cells, spin and carefully remove SN
12. Resus in 620uL cold BD Sample Buffer

Cell Viability Stain

1. Add 3.1 µL of 2 mM Calcein AM and 3.1 µL of 0.3 mM DRAQ7™, (1:200 dilution) gently pipette mix
2. Incubate IN DARK @37°C
3. Strain cells thru flow tube cap and immediately count
    a. Keep cells in dark on ice!

Cell Count

1. Rhapsody pw: Rh@ps0dy
2. Rhapsody scanner: 10uL each side of INCYTO slide
3. Rhapsody: select Hemocytometer, scan side A, side B
    a. Experiment name: C#
    b. Sample name: eoe dupi
    c. User: initials
4. Tap Analysis to view counts: cells/uL
    a. **should be around 322 cells/uL**

Prep:

1. Set Thermomixer to 42°C
2. cDNA and TCR/BCR kit thaw on ice

Cartridge Prime & Treat:

1. Front slider open and check waste, load new 5mL tube
2. Prime cartridge and leave for 1 min RT

Step	Volume (µL)	Pipette mode
100% ethyl alcohol	700	Prime/Treat[a]
Air	700	Prime/Treat
Cartridge Wash Buffer 1 (PN 650000060)	700	Prime/Treat

3. Treat cartridge and use within 4 hours

Step	Volume (µL)	Pipette mode
Air	700	Prime/Treat
Cartridge Wash Buffer 1 (PN 650000060)	700	Prime/Treat
Leave the cartridge on the tray at room temperature for 10 minutes.		
Air	700	Prime/Treat
Cartridge Wash Buffer 2 (PN 650000061)	700	Prime/Treat

Thaw

cDNA kit:	● RT buffer ● dNTP ● DT 0.1 M DTT ● RNase Inhibitor ● Reverse Transcriptase ● H2O ● Bead RT/PCR Enhancer
TCR/BCR kit:	● TSO ● MgCl2 ● Elution Buffer
Cartridge Reagent kit:	● Sample Buffer ● 1 M DTT ● Lysis Buffer ● Enhanced Cell Capture Beads ● Bead Wash buffer ● Stain Buffer ● **CW1 & 2 leave @RT for <=30 mins**
Viability Markers:	- 2mM Calcein AM (from stock 1mg + 503uLDMSO) - 0.3mM DRAX7 ***Light sensitive, do not thaw until needed**

Prep

RBC Removal:	- EasySep RBC Depletion Reagent (4°C fridge) - Easy Eights Magnet - 6mM EDTA (RT) - Lymphopop (RT) - DPBS (Ca,Mg free) + 2% BSA/FBS

Antibody Cocktails

Ab cocktail-1	uL	Ab cocktail-2	uL
FceRIa	1	**Siglec 8**	0.5
CD16	1	**CD66b**	0.5
CD32	1	**AbSeq Abs**	1/sample
CD64	1	**Fc block**	5/sample
Staining Buff	196	**Staining Buff**	Fill to 180, 200 if not multiplexing
Total	**200**	**Total**	**180 or 200**

Rhapsody Cart Load & Start

1. Minimize time between cell pool and single cell capture
2. Calculate pooling volumes for each sample
3. Pipette mix gently then pool. Add Sample Buffer up to 650uL
4. With cartridge on Express, Load Cells:

Pipette	Material	Volume (uL)	Pipette Mode
P1200M	Air	700	Prime/Treat
Set to Cell Load mode, manually pipette mix cells			
P1200M	Pooled cells	575	**Cell Load**

5. Ensure cartridge surface if clean for optimal scanning
6. Move to Rhapsody scanner to Incubate @ RT 15mins
   a. Enter time delay 15 mins > Start Cell Load Scan
   b. Prep Cell Capture Beads: Place on magnet, remove SN, add 750uL cold sample buffer, pipette mix. Keep on ice
   c. Prepare Lysis buffer: add 75uL of 1M DTT to 15mL Lysis buffer bottle, check box on label. Invert and keep on ice
7. **Scan for cell load**, check images on Rhapsody file browser
8. When scan complete, Eject cart and confirm analysis is running

Load Cell Capture Beads

1. Place cart on Express
2. **Bead Load**:

Pipet	Material	uL	Pipet Mode
P1200M	Air	700	Prime/Treat
Set to Bead Load mode, pipet will draw up 40uL air fist. Manually pipette mix cells first			
P1200M	Pooled cells	575	**Bead Load**

3. Incubate cart @RT 3mins
4. Carefully move cart to Rhapsody
5. Run Bead Load Scan
6. **Bead Wash**:

Pipet	Material	uL	Pipet Mode
P1200M	Air	700	Wash
P1200M	Cold Sample Buffer	700	Wash
P1200M	Air	700	Wash
P1200M	Cold Sample Buffer	700	Wash

7. Move cart to Rhapsody and run **Bead Wash scan**
   a. To check scan images: Program Files > Experiments > C# > …. > Bead Wash

$$\mathrm{V} = NP\frac{1.36}{\mathrm{C}}$$

$$P = \frac{1}{\mathrm{n}} \quad n = \#\, samples$$

$$V_A + V_B + V_C + \cdots < 620uL$$

Top off to 650uL with Sample Buffer

$$N = 9{,}000 \text{ x } \#\text{samples}$$

Cartridge max 40,000

Lyse Cells & Bead Retrieval

1. Move cart to Express, move left slider to LYSIS

Pipette	Material	Volume (uL)	Pipette Mode
P1200M	Lysis Buffer + DTT	550	Lysis

2. Incubate @RT 2 mins. Maintain recommended lysis time for best performance
    a. move front slider to BEADS and set P5000M to retrieval mode
3. IMMEDIATELY move side slider into RETRIEVAL **for 30 sec**
    a. Towards end of 30sec take up 5,000uL Lysis Buffer with P5000
4. After 30s, immediately move left slider to middle position (0) and load 4950uL Lysis Buffer
5. Move front slider to OPEN remove 5mL tube
    a. Perform Scanner step: Retrieval
    b. Proceed to Wash Cell Capture Beads
6. Dispose cart, waste container. Wipe down Express

Wash Cell Capture Beads

1. Place 5mL tube on magnet for 1 min. Remove 4mL SN (leave 1mL)
2. Gently pipet mix and transfer to 1.5mL LoBind tube
    a. Can rinse with small volume Lysis Buffer
3. Place on magnet for ≤2 mins. Remove all SN and avoid bubbles
4. Off magnet, add 1mL cold Bead Wash Buffer and mix
5. Repeat wash for total of 2 cycles. Keep in buffer 2nd time on ice until cDNA MM ready
6. Place on ice, start reverse transcription ≤30mins after wash

cDNA Synthesis & Template Switch

1. Set thermomixer to 42°C
2. Make **cDNA/template switching MM keep on ICE**

cDNA/temp switch mix	1X+10% overage (uL)
RT Buffer	44
dNTP	22
RT 0.1 M DTT	11
Bead RT/PCR Enhancer	13.2
Rnase Inhibitor	11
Reverse Transcriptase	11
Nuclase-free H2O	107.8
Total:	**220**

3. Place 1.5mL beads on magnet >2 mins. Remove SN.
4. Add 200uL cDNA MM. <u>Pipet mix and transfer to new tube</u>
5. Incubate on thermomixer 1200rpm @42°C for 30mins
    a. During last 2mins make **TSO mix**:
    b. 6.6uL TSO + 2.2uL 1 M MgCl2
    c. Vortex briefly, keep on ice
6. Once 30mins is up, add 8uL TSO mix and gently pipet mix.
7. Incubate on thermomixer again 1200rpm @42°C for 30mins
8. Place on magnet until clear, discard SN
9. Remove from magnet and gently resus in 75uL Elution Buffer. DO NOT VORTEX!

Stopping point: can store Capture Beads up to 4-7 days at 2-8°C
DO NOT FREEZE CAPTURE BEADS

**Check Rhapsody scanner images and metrics. Transfer files to Google Drive

Day 2: PCR1, RPE

Denaturation & Self-Hybridization

1. With Day 1 beads, pipet mix to resus
2. Incubate @95 °C in a heat block 5 mins with blue cap to secure lid
    a. After 5mins crack cap to release air pressure
    b. Change heat block to 80°C
3. Move fast!
    a. Spin down, on magnet for ~30sec until clear
    b. IMMEDIATELY transfer ALL SN to new tube
4. Keep at 4°C until ready for PCR1
5. Resus beads in 1.5mL Hybridization Buffer. Do not vortex!
6. Incubate @1200rpm and 25C for 2mins
    a. Change thermomixer to 37°C
7. Spin down briefly. Carefully open lid, if droplets on cap, pipet transfer into tube

TCR/BCR Extension (Beads)

1. Make TCR/BCR Extension Mix. Gently vortex. **Keep @RT**

Reagent	1X+5% overage (uL)
TCR/BCR Extension Buffer	21
dNTP	21
TCR/BCR Extension Enzyme	10.5
Nuclease-Free Water	157.5
Total	**210**

2. Place beads tube on magnet for <2mins. Remove SN
3. Resuspend with 200uL of TCR/BCR extension mix. Pipet mix
4. Incubate in thermomixer @1200rpm 37°C 30 mins
    a. Can start Abseq/ST PCR1
5. Spin beads down briefly and keep on ice

TCR/BCR Exonuclease I Treatment

1. Make Exonuclease mix. Gently vortex & spin. **Keep @RT**
    a. Keep Exonuclease I in freezer until needed

Exonuclease I Mix	1X+5% overage (uL)
10X Exonuclease Buffer	21
Exonuclease I	10.5
Nuclease-Free Water	178.5
Total	**210**

2. Place beads tube on magnet for <1min. Remove SN
3. Remove from magnet, resus in 200uL Exonuclease I Mix
4. Incubate in thermomixer @1200rpm 37C for 30mins
5. Incubate on heat block (no shaking) @80°C for 20 mins
    a. Change heat block to 95°C
6. After 20mins place beads tube on ice for ~1min
7. Spin down and place on magnet for <1min. Remove SN
8. Continue to WTA RPE, resus beads in 174uL RPE mix **or if stopping,** resus in 200uL cold Bead Resuspension Buffer

STOPPING POINT: can store exonuclease treated TCR/BCR beads up to 3 months at 2-8°C

Prep:

1. PCR Cleanup beads from 2-8°C to RT @ least 30mins
2. Set Thermomixer to 25°C
3. Set heat block to 95°C
4. Make 80% ethanol
5. Thaw WTA kit

WTA Kit	dNTP
	H20
	PCR MM
	Universal Oligo
	Sample Tag PCR1 Primer
	BD Abseq Primer
	Bead RT/PCR Enhancer
	WTA Extension Buffer
	WTA Extension Primers
TCR/BCR Kit	Elution Buffer
	Hybridization Buffer
	TCR/BCR Buffer
	Bead Resuspension Buffer
cDNA kit	10X Exonuclease 1 Buffer

Supernatant PCR1 (Abseq/Sample Tag):

1. Abseq/ST PCR1 mix. Gently vortex & spin. Keep on ice

Abseq/ST PCR1 Mix	1X+10% overage
PCR MM	110
Universal Oligo	11
Sample Tag PCR1 Primer	1.1 (if no SMK replace with water)
BD Abseq Primer	11
Nuclease-free H2O	13.2
Total:	**146.3**

2. 133uL Abseq/ST PCR1 MM + 67uL SN (Abseq/ST product) in new 1.5mL. Pipet mix 10x. Do NOT vortex
3. Aliquot 50uL into 4 PCR strip tubes. (any residual add to any tube)
4. Thermocycler run ~1hr 10mins:

Abseq/ST PCR1	Cycles	Temp °C	Time
Hot Start	1	95	3 min
Denaturization	10-11	95	30 sec
Annealing		60	30 sec
Extension		72	1 min
Final extension	1	72	5 min
Hold	1	4	∞

# of cells	Recommended Cycles
7,500->20,000	11
20,000	10

5. Briefly spin down, pool in new 1.5mL tube. Keep on ice until ready for purification

RPE of Beads with cDNA

1. Make RPE (Random Priming & Extension) mix. **Keep @RT**

RPE mix	1X+5% overage (uL)
WTA Extension Buffer	21
WTA Extension Primers	21
Nuclease-Free Water	140.7
Total	**182.7**

Use all beads, sub-sampling not recommended for our combination of libraries and assays

2. Pipet mix Exonuclease-treated beads, on magnet ~2 mins, Remove SN
3. Spin down and back on magnet for ~2min. Remove SN
4. Take off magnet. Resus in 174uL RPE mix. Pipet mix 10x
5. **Incubation cycle (manually change thermomixer):**
   a. 95°C in heat block 5mins
   b. Thermomixer @1200rpm 37°C for 5mins
   c. Thermomixer @1200rpm 25°C for 15mins
   d. Keep @RT
   e. **Take sample out, set thermomixer to 25°C**
6. Make Extension enzyme mix. Keep @RT
   a. Keep WTA Extension Enzyme in fridge until needed

Extension Enzyme mix	1X+5% overage (uL)
dNTP	8.4
Bead RT/PCR Enhancer	12.6
WTA Extension Enzyme	6.3
Total	**27.3**

7. Add 26uL of Extension Enzyme mix into 174uL of beads (for total of 200uL). Keep @RT until ready for incubation
8. Thermomixer run ~55mins
   a. Ramp rates set at MAX and program set to TIME CONTROL (not temp control)

RPE Program	Temp °C	Time
1200rpm	25	10 mins
1200rpm	37	15 mins
1200rpm	45	10 mins
1200rpm	55	10 mins

9. Brief spin and place on magnet until clear. Remove SN
10. Take off magnet and resus in 205uL Elution Buffer
11. Denature RPE product off beads with heat block 95°C 5 mins. **Do NOT go over 5 mins.** Use blue cap guard. Carefully crack open after incubation to release pressure
12. On magnet until clear. Immediately transfer 300uL SN into new 1.5mL tube and keep @RT. <u>This is WTA RPE Product</u>
13. Continue to T/BCR PCR1, resus beads in 200uL TCR/BCR PCR1 mix **<u>or if stopping,</u>** resus beads with 200uL cold Bead Resuspension Buffer. Store @4°C or on ice. <u>This is TCR/BCR library</u>

If reamplifying WTA off beads (extra wash step)

1. With capture beads in bead resus buffer, magnet and remove SN
2. Resus in 200uL EB
3. Place on heat block 95 degrees for **1 min**,
4. Magnet and remove SN
5. Beads ready for second round of RPE

Abseq/ST PCR1 Purification

1. Vortex AMPure XP magnetic beads for 1 min
2. Check Abseq/ST PCR1 is @200uL, if not add H2O
3. **Add 280uL AMPure beads**
4. Pipet mix 10x and incubate 5 mins
5. Place on magnet 5 mins. Remove SN
6. Gently add <u>500uL</u> 80% ethanol. Wait 30sec and remove. Repeat for total of 2 washes
7. Use small volume pipet to remove SN
8. Air dry for 5mins. Place in fume hood to acc. Check dryness by looking for bead glossiness
9. Take off magnet and resus with **50uL Elution Buffer**
   Will be clumpy. Vigorously pipet. Small clumps ok
10. Incubate 2mins @RT
11. Spin down briefly and place on magnet until clear.
12. Pipette ~50uL eluate into new 1.5mL tube.

STOPPING POINT: can store Abseq/ST PCR1 for 24hrs at 2-8°C or -25 to -15°C for up to 6 months

<u>If not multiplexing can skip ST PCR2</u>

1. Make Sample Tag PCR2 mix. **Keep on ice**

Sample Tag PCR2	1X+10% overage (uL)
PCR MM	27.5
Universal Oligo	2.2
Sample Tag PCR2 primer	3.3
Nuclease-free H2O	16.5
Total	**49.5**

10. Add 5uL ST PCR1 product to 45uL PCR2 MM
11. Gently vortex and spin down
12. Thermocycler TCR & BCR Touchdown and Sample Tag PCR2

ST PCR2	Cycles	°C	Time
Hot Start	1	95	3 min
Denat		95	30 s
Anneal	10-11	60	30 s
Extend		72	1 min
Final Extend	1	72	5 min
Hold	1	4	∞

TCR/BCR PCR1 (beads)

1. Set Thermocycler TCR/BCR PCR1 program Hot start at 95°C
2. Make TCR/BCR PCR1 mix. **Keep on ice**

TCR/BCR PCR1	1X+5% overage (uL)
PCR MM	105
TCR/BCR Universal Oligo N1	10.5
Bead RT/PCR Enhancer	12.6
TCR N1 Primer	2.52
BCR N1 Primer	2.52
Nuclease-free H2O	76.88
Total	**210**

3. Briefly vortex & spin beads. Place on magnet ~1min. Remove SN
4. Take off magnet and resus in 200uL PCR1 mix. Do not vortex!
5. Mix well and aliquot 50uL into 4 PCR strip tubes.
   a. Pipet mix before putting in thermocycler
6. Once thermocycler is @95°C, start program ~1hr 10mins:

TCR/BCR PCR1	Cycles	Temp °C	Time
Hot Start	1	95	3 min
Denaturization	10-11	95	30 sec
Annealing		60	1 min
Extension		72	1 min
Final extension	1	72	5 min
Hold	1	4	∞

7. Briefly spin down PCR tubes and pool in new 1.5mL tube
8. Place on magnet until clear. **Transfer ~200uL SN into new 1.5mL**. This is TCR/BCR PCR1 Product
9. Resus beads in 200uL Cold Bead Resuspension Buffer. Do not vortex. Labe as TCR/BCR PCR1 Beads and store 4-8°C
   a. Enhanced Cell Capture Beads that have been through TCR/BCR extension and WTA RPE can be kept as backup for up to 4 months. Can be used to reamplify for TCR/BCR/WTA though products may contain more noise from previous primers

TCR/BCR PCR1 Purification

1. Vortex AMPure XP magnetic beads for 1 min
2. Check TCR/BCR PCR1 is @200uL, if not add H2O
3. **Add 140uL AMPure beads**
4. Pipet mix 10x and incubate 5mins
5. Place on magnet 5 mins. Remove SN
6. Gently add 500uL 80% ethanol. Wait 30sec and remove. Repeat for total of 2 washes
7. Use small volume pipet to remove SN
8. Air dry for 5mins. Check dryness
9. Take off magnet and resus with **50uL Elution Buffer** Will be clumpy. Vigorously pipet. Small clumps ok
10. Incubate 2mins @RT
11. Spin down briefly and place on magnet until clear.
12. Pipette ~50uL eluate into new 1.5mL tube.

STOPPING POINT: can store TCR/BCR PCR1 for 24hrs at 2-8°C or -15°C t0 -25°C up to 6 months

Can proceed to Touchdown PCR2 for TCR&BCR and ST then purify (2hrs)

RPE (SN) Purification

1. Vortex AMPure XP magnetic beads for 1 min
2. Check WAT RPE is @200uL, if not add H2O
3. **Add 320uL AMPure beads**
4. Pipet mix 10x and incubate 10 mins
5. Place on magnet 5 mins. Remove SN
6. Gently add 1mL 80% ethanol. Wait 30sec and remove. Repeat for total of 2 washes
7. Use small volume pipet to remove SN
8. Air dry for 5mins. Place in fume hood to acc. Check dryness by looking for bead glossiness
9. Take off magnet and resus with **40uL Elution Buffer**
10. Incubate 2mins @RT
11. Spin down briefly and place on magnet until clear.
12. Pipette ~40uL eluate into new PCR tube

WTA RPE PCR (SN)

1. Make WTA RPE PCR mix

WTA RPE PCR mix	1X+5% overage (uL)
PCR MM	63
Universal Oligo	10.5
WTA Amplification Primer	10.5
Total	**84**

2. Add 80uL RPE PCR mix to tube with 40uL of purified RPE product. Pipet mix 10x
3. Split into 2 PCR tubes 60uL each
4. Run Thermocycler WTA RPE PCR program ~1h

WTA RPE PCR	Cycles	Temp	Time
Hot Start	1	95°C	3 min
Denaturization	7,500 cells: 13 20,000 cells: 11	95°C	30 sec
Annealing		60°C	1 min
Extension		72°C	1 min
Final extension	1	72°C	2 min
Hold	1	4°C	∞

WTA RPE PCR Purification (single sided cleanup)

1. Spin down PCR tubes and pool, can keep in PCR tubes
2. Vortex AMPure XP magnetic beads for 1 min
3. Check WTA RPE is 120uL, if not add H2O
4. **Add 96uL AMPure beads**
5. Pipet mix 10x and incubate 5 mins
6. Place on magnet 5 mins. Remove SN
7. Gently add 200uL 80% ethanol. Wait 30sec and remove. Repeat for total of 2 washes
8. Use small volume pipet to remove SN
9. Air dry for 5mins. Place in fume hood to acc. Check dryness by looking for bead glossiness
10. Take off magnet and resus with **40uL Elution Buffer**
11. Incubate 2mins @RT
12. Spin down briefly and place on magnet until clear.
13. Pipette ~40uL eluate into new 1.5mL tube.

STOPPING POINT: can store WTA RPE product for 6 weeks at 2-8°C or -15°C t0 -25°C up to 6 months

Day 3: PCR2, index PCR, Qubit and Tape Station

TCR/BCR & Sample Tag Touchdown PCR2 of PCR1 Product

13. Make a TCR and a BCR PCR2 mix. **Keep on ice**

TCR and BCR PCR2	1x+10% overage (uL)
PCR MM	27.5
TCR/BCR Universal Oligo N2	2.2
TCR **or** BCR N2 primer	6.6
Nuclease-free H2O	13.2
Total	**49.5**

14. Make Sample Tag PCR2 mix. **Keep on ice**

Sample Tag PCR2	1X+10% overage (uL)
PCR MM	27.5
Universal Oligo	2.2
Sample Tag PCR2 primer	3.3
Nuclease-free H2O	16.5
Total	**49.5**

15. Add 5uL PCR1 product to 45uL PCR2 MM for ST, TCR, BCR
16. Gently vortex and spin down
17. Thermocycler TCR & BCR Touchdown and Sample Tag PCR2

TCR&BCR PCR2	Cycles	°C	Time
Hot Start	1	95	3 min
Denat I	15	95	30 s
Anneal I		75-60	1 min
Extend I		72	1 min
Denat II	8	95	30 s
Anneal II		60	1 min
Extend II		72	1 min
Final extend	1	72	5 min
Hold	1	4	∞

TCR&BCR & ST PCR2 Purification

1. Spin down PCR tubes, **check if volume is 50uL**
2. Vortex beads ~1min. To 50uL of PCR2 product add:
   a. **TCR & BCR: Add 35uL AMPure beads each**
   b. **ST: Add 60uL AMPure beads**
3. Pipet mix 10x and incubate 5 mins
4. Place on strip tube magnet 3 mins. Remove SN
5. Gently add 200uL 80% ethanol. Wait 30sec and remove. Repeat for total of 2 washes
6. Use small volume pipet to remove SN
7. Air dry for 1min. Check dryness
8. Take off magnet and resus with **30uL Elution Buffer** Will be clumpy. Vigorously pipet. Small clumps ok
9. Incubate 2mins @RT
10. Spin down briefly and place on magnet until clear.
11. Pipette ~50uL eluate into new 1.5mL tubes

STOPPING POINT: can store TCR and BCR PCR2 products for 24hrs at 2-8°C or -25 to -15°C for up to 6 months

Qubit & TapeStation: PCR1 and PCR2 Products

1. Qubit expected concentration **~0.5-10 ng/uL**
   a. Standard 1&2: 10uL standard + 190uL Qubit Buffer
   b. Sample: 2uL + 198uL Qubit Buffer
   c. Incubate in dark for 2mins
2. Make dilutions for TapeStation 1 – 1.3 ng/uL:
   a. **Abseq&ST:** 5-8uL for TS and index PCR
   b. **WTA RPE:** 2uL for TS then dilute to ~2nM

Prep:

1. Make 80% ethyl alcohol
2. Ampure beads to RT
3. Equilibrate @RT:

Qubit	Qubit Buffer
	Standard #1
	Standard #2
Tape Station	D5000 Screen Tape
	D5000 Ladder
	D5000 Sample Buffer

4. Thaw:

WTA kit	PCR MM
	Universal oligo
	dNTP
	Sample Tag PCR primer
	Assigned Forward Primer
	Reverse Primers 1-4
	Nuclease-free H2O
	Elution Buffer
Tape Station	TCR/BCR universal oligo
	TCR N2 Primer
	BCR 2 Primer
	TCR/BCR Extension Buffer
	TCR/BCR Extension Primer

Sample and Library Tag Format

...	C10	C11	...	...
Forward	1	2	3	4
WTA	1	1	1	1
AbSeq	2	2	2	2
ST	3	3	3	3
TCR	4	4	4	4
BCR	5	5	5	5

TCR&BCR RPE of PCR2 Product (random primer extension)

1. Make 2x TCR & BCR Random Primer mix. **Keep @RT**

TCR&BCR RP mix	1x	2x+10% overage (uL)
TCR/BCR Extension Buffer	5	11
TCR/BCR Extension Primer	2.5	5.5
Nuclease-free H2O	34	74.8
Total	**41.5**	**91.3**

2. Add 41.5uL of RP mix + 5uL TCR&BCR PCR2 at ~1ng/uL
3. Thermocycler program:

TCR&BCR RP	Cycles	Temp	Time
Denaturization/ Random Priming	1	72°C	5 min
		37°C	5 min
		25°C	15 min

4. During RP run make Primer Extension mix. **Keep @RT**

TCR&BCR Extension mix	1x	2x+10% overage (uL)
dNTP	2	4.4
TCR/BCR Extension Enzyme	1.5	3.3
Total	**3.5**	**7.7**

5. When RP run is done briefly centrifuge tubes, add 3.5uL Extension mix to each RP product tube for total of 50uL
6. Run thermocycler extension program

TCR&BCR Extension	Cycles	Temp	Time
Extension	1	25°C	10 min
		37°C	15 min
		45°C	10 min
		55°C	10 min

TCR & BCR RPE Purification

1. Spin down PCR tubes. Do not pool!
2. **Check volume is 50uL**
3. Vortex beads ~1min. **Add 90uL AMPure beads**
4. Pipet mix 10x and incubate 5 mins
5. Place on strip tube magnet 3mins. Remove SN
6. Gently add <u>200uL</u> 80% ethanol. Wait 30sec and remove. Repeat for total of 2 washes
7. Use small volume pipet to remove SN
8. Air dry for 1min. Can place in fume hood to acc. Check dryness by looking for bead glossiness
9. Take off magnet and resus with **30uL or 21uL Elution Buffer**
10. Incubate 2mins @RT
11. Spin down briefly and place on magnet until clear.
12. **Pipette 21uL eluate into new PCR tube for TCR&BCR index PCR**

Abseq & Sample Tag Index PCR

1. Make 2x TCR & BCR Random Primer mix. **Keep on ice**

Abseq/ST index PCR mix	1x	1x+10% overage (uL)
PCR MM	25	27.5
Library For Primer	2	2.2
Library Rev Primer 2 or 3	2	2.2
Nuclease-free H2O	18	719.8
Total	**47**	**49.35**

2. With Abseq & ST PCR2 dilution @0.1-1.1ng/uL, in new PCR strip tube:
   a. 47uL <u>index 2 PCR</u> MM + 3uL Abseq PCR1 dilution
   b. 47uL <u>index 3 PCR</u> MM + 3uL ST PCR2 dilution
3. Gently vortex and spin down, run thermocycler:

Abseq & ST index PCR	Cycles	°C	Time
Hot Start	1	95	3 min
Denat	Refer to table below	95	30 s
Anneal		60	30 s
Extend		72	30 s
Final Extend	1	72	1 min
Hold	1	4	∞

Conc. Index PCR input (ng/uL)	Recommended # Cycles
0.5-1.1	6
0.25-0.5	7
0.1-0.25	8

Abseq & ST index PCR Purification

1. Spin down PCR tubes
2. **Check volume is 50uL**
3. Vortex beads ~1min. **Add 40uL AMPure Beads**
4. Pipet mix 10x and incubate 5 mins
5. Place on strip tube magnet 5mins. Remove SN
6. Gently add <u>200uL</u> 80% ethanol. Wait 30sec and remove. Repeat for total of 2 washes
7. Use small volume pipet to remove SN
8. Air dry for 1min. Check dryness
9. Take off magnet and resus with **50uL Elution Buffer**
10. Incubate 2mins @RT
11. Spin down briefly and place on magnet until clear.
12. Pipette ~50uL eluate into new 1.5mL tube
13. <u>Ready for Qubit & Tapestation</u>

STOPPING POINT: can store Abseq & ST final product for 24hrs at 2-8°C or -15°C t0 -25°C up to 6 months

TCR & BCR index PCR of RPE Product

1. Make 2x TCR & BCR index PCR mix. **Keep on ice**

TCR&BCR index PCR mix	1x	1x+10% overage (uL)
PCR MM	25	27.5
Library Forward Primer	2	2.2
Library Rev Primers 4 or 5	2	2.2
Total	**29**	**30.45**

2. In new PCR strip tube:
   - 29uL index 4 MM + 21uL of TCR purified product
   - 29uL index 5 MM + 21uL of BCR purified product
3. Gently vortex and spin down
4. Run thermocycler TCR&BCR Index PCR

TCR&BCR index PCR	Cycles	Temp	Time
Hot Start	1	95°C	3 min
Denaturization	10-11* typically low concentration so we changed to 11	95°C	30 sec
Annealing		60°C	30 sec
Extension		72°C	30 sec
Final Extension	1	72°C	1 min
Hold	1	4°C	∞

TCR & BCR index PCR Purification

1. Spin down PCR tubes. Transfer 50uL of TCR and BCR index PCR product into new PCR tubes
2. **Vortex beads ~1min. Add 32.5uL AMPure Beads**
3. Pipet mix 10x and incubate 5 mins
4. Place on strip tube magnet 5mins. Remove SN
5. Gently add 200uL 80% ethanol. Wait 30sec and remove. Repeat for total of 2 washes
6. Use small volume pipet to remove SN
7. Air dry for 1min. Check dryness
8. Take off magnet and resus with **30uL Elution Buffer**
9. Incubate 2mins @RT
10. Spin down briefly and place on magnet until clear.
11. Pipette ~50uL eluate into new 1.5mL tube
12. Ready for Qubit & Tapestation

STOPPING POINT: can store TCR & BCR final product for 24hrs at 2-8°C or -15°C t0 -25°C up to 6 months

Beads:Sample = 0.65

If using 50uL T/B index PCR product: add 32.5uL beads

If using 40uL T/B index PCR product: add 26uL beads (og protocol)

WTA RPE index PCR of purified RPE PCR product

1. Make WTA RPE index PCR mix

WTA RPE index PCR mix	1X+10% overage (uL)
PCR MM	27.5
Library Forward Primer	5.5
Library Reverse Primer 1	5.5
Nuclease-free H2O	5.5
Total	**44**

2. Dilute WTA RPE product concentration to 2nM. If concentration <2nM do not dilute.
3. In PCR strip tube add 40uL MM + 10uL 2nM WTA
4. Pipet mix 10x and program Thermocycler:

WTA index PCR	Cycles	Temp	Time
Hot Start	1	95°C	3 min
Denaturization	*table below	95°C	30 s
Annealing		60°C	30 s
Extension		72°C	30 s
Final extension	1	72°C	1 min
Hold	1	4°C	∞

Concentration of diluted WTA RPE	Recommended Cycles
<2nM	9
2nM	8

WTA index PCR Purification (single sided)

1. Spin down PCR tubes
2. **Add 60uL H2O for final product volume 110uL**
3. Transfer 100uL to new PCR tube
4. **Vortex beads ~1min. Add 65uL AMPure Beads**
5. Pipet mix 10x and incubate 5 mins
6. Place on strip tube magnet 5mins. Remove SN
7. Gently add 200uL 80% ethanol. Wait 30sec and remove. Repeat for total of 2 washes
8. Use small volume pipet to remove SN
9. Air dry for 1min. Check dryness. **Do not overdry.**
10. Take off magnet and resus with **30uL Elution Buffer**
11. Incubate 2mins @RT
12. Spin down briefly and place on magnet until clear.
13. Pipette ~30uL eluate into new 1.5mL tube
14. Ready for Qubit and Tapestation

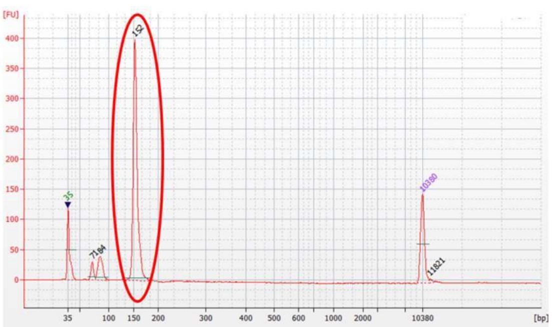
STOPPING POINT: can store WTA final product at -15°C to -25°C up to 6 months

Qubit & TapeStation

1. Qubit expected concentration **~0.5-10 ng/uL**
   a. Standard 1&2: 10uL standard + 190uL Qubit Buffer
   b. Sample: 2uL + 198uL Qubit Buffer
   c. Incubate in dark for 2mins
2. Use Qubit values to get dilutions of 2uL of 1ng/uL for Tapestation
   a. Dilute with Nuclease-free water
3. Tapestation load and label strip tubes:
   a. Ladder: 2uL Ladder + 2uL Sample Buffer
   b. Samples: 2uL Sample + 2uL Sample Buffer

Tapestation Trace Files:

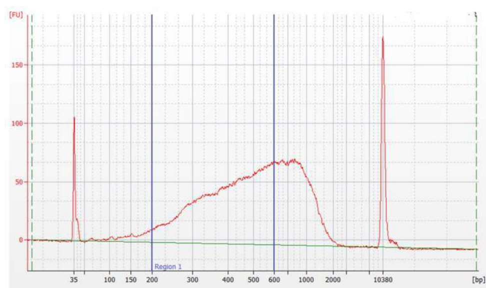
Figure 2 Sample Bioanalyzer High Sensitivity DNA trace - AbSeq/Sample Tag PCR1



	Size [bp]	Conc. [pg/µl]	Molarity [pmol/l]	Observations
	35	125.00	5,411.3	Lower Marker
2	71	51.24	1,087.6	
3	84	123.99	2,225.8	
4	152	838.65	8,334.2	
5	10,380	75.00	10.9	Upper Marker
6	11,821	0.00	0.0	

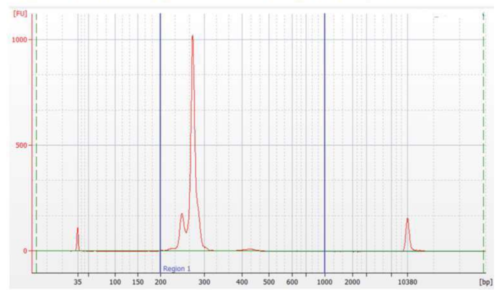
RPE PCR product

Figure 1 Sample Bioanalyzer High Sensitivity DNA trace - RPE PCR product trace



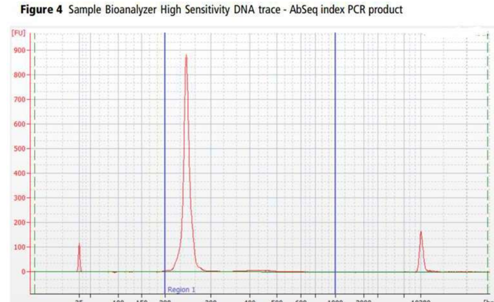
Sample Tag index PCR product

Figure 3 Sample Bioanalyzer High Sensitivity DNA trace - Sample Tag index PCR product



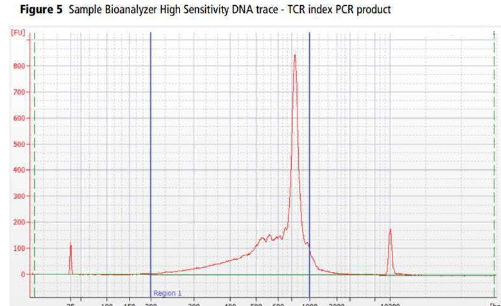
Abseq index PCR product

Figure 4 Sample Bioanalyzer High Sensitivity DNA trace - AbSeq index PCR product



TCR index PCR product

Figure 5 Sample Bioanalyzer High Sensitivity DNA trace - TCR index PCR product



BCR index PCR product

Should be double peak (left shoulder)

Figure 6 Sample Bioanalyzer High Sensitivity DNA trace - BCR index PCR product